

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1035

Fudge Hollow 138 kV
September 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1035. The Interconnection Customer proposes a 71.1 MW solar photovoltaic facility interconnecting at the Fudge Hollow 138 kV substation in the Virginia area. The request was received on July 28, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 269 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,845,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Talley Road-Lowmoor 138kV L	Aspen-Lowmoor Substation 345kV	109.7	230	9.8
Clifton Forge-Talley Road 138kV line	Lowmoor-Aspen 345kV	109.0	226	9.8
Richmond Q-Clifton Forge 138kV Line	Lowmoor-Aspen 345kV line	107.6	219	9.8
Weyers Cave-Greenbank 138kV	Crandall-Dayton 345kV	106.6	398	7.6
Richmond Q-Kattelville 138kV line	Lowmoor-Aspen 345kV	106.1	212	9.8
Michigan Power Lp-Osceola 138kV	Lowmoor-Aspen 345kV	105.6	355	13.9
Osceola Substation-Lone Tree 138kV	Lowmoor-Aspen 345kV	105.3	361	13.9
Yucaipa-Craigsville Substation 138kV	Lowmoor-Aspen 345kV	102.0	205	9.8
Redland-Carodean 138kV Line	Dayton Substation-Crandall 345kV	101.9	368	7.9
Kattelville-Craigsville 138kV Line	Aspen-Lowmoor 345kV	101.8	216	9.8
Verona-Greenbank 138kV	Dayton-Crandall 345kV	100.8	288	5.5
Crandall-Carodean 138kV	Crandall-Dayton 345kV	100.7	400	7.9
Redland-Verona 138kV	Dayton-Crandall 345kV	100.4	298	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Talley Road-Lomoor Substation 138kV	Reconductor line	\$3,240,000	22% (\$725,000)
Clifton Forge-Talley Road 138kV line	Reconductor line	\$2,548,000	22% (\$570,000)
Clifton Forge-Richmond Q 138kV	Reconductor line	\$577,000	22% (\$129,000)
Weyers Cave-Greenbank 138kV line	Reconductor line	\$940,000	10% (\$99,000)
Richmond Q-Kattelville 138kV line	Reconductor line	\$10,251,000	22% (\$2,294,000)
Michigan Power Lp-Osceola 138kV L	Reconductor line	\$3,537,000	4% (\$138,000)
Oceola-Lone Tree 138kV	Reconductor line	\$1,782,000	4% (\$69,000)
Craigsville-Yucaipa 138kV	Reconductor line	\$1,943,000	22% (\$435,000)
Redland-Carodean 138kV Line	Reconductor line	\$268,000	9% (\$23,000)
Craigsville-Kattelville 138kV	Reconductor line	\$3,729,000	22% (\$835,000)
Verona-Greenbank 138kV	Reconductor line	\$970,000	12% (\$114,000)
Crandall-Carodan 138kV line	Reconductor line	\$1,190,000	9% (\$102,000)
Redland-Verona 138kV	Reconductor line	\$2,655,000	12% (\$312,000)
	Total allocated to project		\$5,845,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.